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Haneeka

RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

B.Sc (General) Degree in Applied Sciences
First Year – Semester I Examination – September / October 2019

CHE 1201 – GENERAL CHEMISTRY

Time: Two (2) hours

1eV = 1.602×10^{-19} J, Planck's constant = 6.626×10^{-34} J s, electron charge = 1.602×10^{-19} C
Speed of light = 2.99×10^{8} m s^{-1}
Use of nonprogrammable calculator is permitted

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1. a) Bohr's atomic model solves the limitations of Rutherford atomic model. Explain. (20 marks)

b) For a hydrogen atom, show that
   i) Bohr radius (r) = $n^2 \times$ constant and
   ii) Total energy (E_T) in the n^{th} orbital $= -\left(\frac{1}{n^2}\right) \times$ constant

The force of attraction between charges (F) $= \frac{q_1 q_2}{4\pi\varepsilon_0 r^2}$, Centripetal force $(a) = \frac{mv^2}{r}$

Potential and kinetic energy of an electron in an orbit are $-\frac{e^2}{4\pi\varepsilon_0 r}$, and $\frac{1}{2}mv^2$ respectively. (30 marks)

c) Explain the origin of the emission spectrum of hydrogen (10 marks)

d) Johan Balmer obtained a good linear relationship as given below, between frequencies of each line in the emission spectrum of hydrogen and $\frac{1}{n^2}$.

$$\nu = -4 \times 8.2202 \times 10^{14} \left(\frac{1}{n^2}\right) + 8.2202 \times 10^{14}$$

Show that the wave number $(\overline{\nu}) = 109680 \left(\frac{1}{2^2} - \frac{1}{n^2}\right)\ cm^{-1}$ (20 marks)

e) Calculate the energy (kg mol^{-1}) required for the ionization of an electron from the ground state of the hydrogen atom. (20 marks)

c) The compound trimethylamine $[(CH_3)_3N]$ and trisilylamine $[(SiH_3)_3N]$ have similar formulae, but totally different structures. Explain the above statement using orbital diagrams.

(30 marks)

d) Write a short note on interhalogen compounds

(20 marks)

5. a) Calculate the lattice energy, U in kJ mol^{-1} for ZnO in the wurtzite structure using the Born-Lande equation and using a Born-Haber cycle. Compare the two answers and comment on it.

$$U = -\frac{-N_A A Z^2 e^2}{4\pi\varepsilon_0 r_0}\left(1 - \frac{1}{n}\right)$$

Hint: $r_0 = 1.99 \times 10^{-10}$ m, the Born exponent, n = 8, and A (Madelung constant) = 1.641 for the wurtzite structure, and Z = charge on the ion.

	$H^{\ominus}$ / kJ mol^{-1}
Energy of sublimation for Zn(s)	130.4
1st ionization energy for Zn(g)	418.61
2nd ionization energy for Zn(g)	1733
Bond dissociation energy for O_2(g)	497
Electron affinity values of oxygen:	
$O(g) + e^- \rightarrow O^-(g)$	141
$O^-(g) + e^- \rightarrow O^{2-}(g)$	- 780
Heat of formation for ZnO(s)	- 350.51

(50 marks)

b) (i). Define the terms polarizability and polarizing power.

(ii). AgCl is insoluble in water whereas AgF is soluble. Explain

(iii). $NaHCO_3$ reacts with an aqueous solution of $FeCl_3$ generating carbon dioxide but not with an aqueous solution of $FeCl_2$. Justify your answer.

(30 marks)

c) The Dipole moment of HCl is 1.11 D, and the distance between the two atoms is 127 pm. Comment on the covalent character of the HCl bond?

(20 marks)

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RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

B.Sc. in Applied Sciences
First year Semester I Examination – March 2021

CHE 1201 – GENERAL CHEMISTRY

Answer any **FOUR** questions only **Time: Two (2) hours**

Planck constant	=	6.626×10^{-34} J s	Velocity of light c	=	2.99×10^{8} m s^{-1}
Avogadro constant	=	6.02×10^{23} mol^{-1}	Rydberg constant R	=	1.09×10^{7} m^{-1}
Mass of the electron	=	9.1×10^{-31} kg	Electron charge e	=	1.60×10^{-19} C
Permittivity of vacuum	=	8.85×10^{-12} kg^{-1} m^{-3} A^{2} (or F m^{-1})			

Use of a non-programmable calculator is permitted.

1. a) State and explain how the fundamental postulates of Bohr theory refined the drawbacks of Rutherford's model of an atom.
(30 marks)

b) Consider an atom whose nucleus has a charge Z_e and an electron moving round the nucleus in an orbit of radius r. Derive a relationship that shows the total energy, $E_T = \left(-\frac{1}{n^2}\right) \times$ constant. Centripetal force $= \frac{MV^2}{r}$, and electrostatic attraction, $F = \frac{Ze^2}{4\pi\varepsilon_0 r^2}$ where Z is the atomic number and ε_0 is the permittivity of free space.
(35 marks)

c) Calculate the ionization energy of hydrogen atom in kJ mol^{-1}
(35 marks)

2. a) State Broglie hypothesis and Heisenberg's uncertainty principle
(20 marks)

b) Calculate the uncertainty in position of an electron moving at a speed of 0.05% of that of light. Is the uncertainty in position larger or smaller than the de Broglie wavelength of the electron?
(15 marks)

c) Briefly discuss the (i) contribution of Max Planck to explain the experimental distribution of energy in the spectrum of a black body and (ii) Ultraviolet Catastrophe
(35 marks)

d) What is meant by photoelectric effect? It requires a photon with a minimum of 6.94 $\times 10^{-34}$ J of energy to eject an electron from a polished Zn surface.

(i) Does electromagnetic radiation with wavelength 210 nm suffice to do this?

...erford atomic model. Explain.

(20 marks)

(ii) If it does, what is the maximum kinetic energy of the electron ejected?

(30 marks)

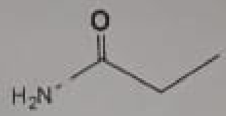
3. a) Draw resonance structures with formal charge on each atom for the cyanate ion OCN^- and comment on the stability of each structure.

(30 marks)

b) In water, acetate anion, CH_3COO^- is more stable than CH_3COOH. Explain

(15 marks)

c) What do you mean by term hybridization? Deduce the hybridization of carbon, nitrogen and oxygen atom in the following molecule.



(30 marks)

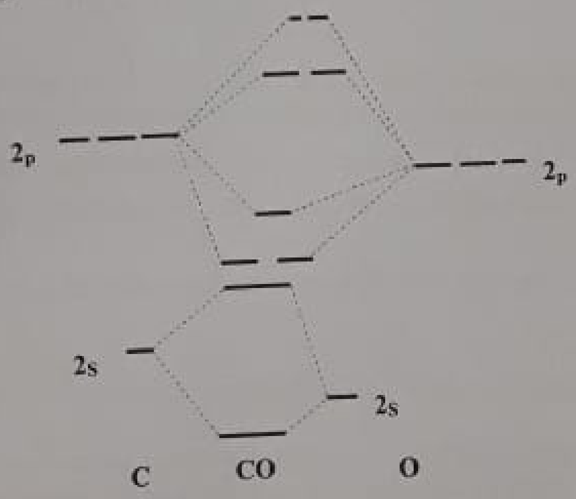
d) The molecular orbital diagram of CO molecule is given below.

(i) Label and insert all electrons present in atomic orbitals and molecular orbitals.

(ii) Determine the bond order of CO molecule.

(iii) Write the electronic configuration of CO molecule.

(25 marks)



4. a) Describe the structural differences between B_2H_6 and C_2H_6. Explain with illustrations.

(30 marks)

b) Be and Mg are both group 2 elements. Among the two elements Mg shows hexahydrated complexes, whereas Be does not. Justify your answer.

(20 marks)

2 a) i. In the quantum-mechanical description. What is the physical significance of the [illegible] square of the wave function, ψ^2? (10 marks)

ii. Show the variation of ψ^2 as a function of distance (r) from the nucleus for 1s and 2[illegible] orbitals (10 marks)

iii. Write four quantum numbers for the last valence electron in a chlorine atom (10 marks)

b) Write short notes on:

i. (A) Ultraviolet catastrophe, (B) Wave-particle duality and (C) De-Broglie wavelength (30 marks)

ii. Describe the Heisenberg's uncertainty principle? (10 marks)

iii. The position of an alpha particle is known with a precision of 0.01 Å. What is the uncertainty involved in the simultaneous measurement of its momentum? (mass of alpha particle = 6.68×10^{-27} kg). (15 marks)

C) A photoelectric surface has a work function of 4 eV. What is the maximum velocity of the photoelectrons emitted by light of frequency $10^{15}\ s^{-1}$ incident on the surface? (15 marks)

3 a) i. Draw possible Lewis structures for (i) ICl_4^- (ii) NO^+ and (iii) SO_4^{2-} and identify the electron geometry and the molecular geometry of each structure. (24 marks)

ii. Identify the hybridization of the central atom of each structure (part i).

b) i. The Lewis structure of O_2 fails to account for the paramagnatism of O_2, but molecular orbital theory correctly predicts its magnetism. Explain with appropriate molecular orbital diagrams. (20 marks)

ii. Write down the molecular orbital electron configuration of O_2 molecule (04 marks)

c) i. According to molecular orbital description, predict the bond order of O_2, O_2^+, and O_2^-

(10 marks)

ii. Arrange above oxygen species in order of increasing bond length (05 marks)

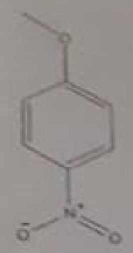
d) Three possible Lewis structures for the thiocyanate ion, NCS^-, are

i. $[N{=}C{=}S]^-$ ii. $[N{\equiv}C{-}S]^-$ iii. $[N{-}C{\equiv}S]^-$

A) Determine the formal charges in each structure. B) Based on the formal charges, which Lewis structure is the dominant one? Explain. (10 marks)

H, H spectrum.

c) Draw all possible resonance forms for the following structure using appropriate arrow notation. Which resonance structure is most stable? and least stable? Draw the resonance hybrid indicating all partial charges.



(15 marks)

4. a) Explain the following:

   (i) AgF is soluble in water while AgI is insoluble in water (20 marks)

   (ii) H_3BO_3 is a very weak acid and direct titration with NaOH is not possible. However, in the presence of organic polyhydroxy compounds such as glycerol, mannitol etc, facilitates the acid–base titration. (20 marks)

   b) Discuss the properties of trimethyl amine $(CH_3)_3N$ and trisilylamine $(SiH_3)_3N$ with respect to their molecular structures. (20 marks)

   c) $SiCl_4$ rapidly hydrolyses in aqueous alkaline media whereas SiF_4 doesn't. Explain with appropriate reactions. (20 marks)

   d) What would you observe if an aqueous solution of Na_2CO_3 is added to an aqueous solution of $AlCl_3$. Explain your answer. (20 marks)

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